

**GRAPH THEORY:
HOMEWORK 5
BOOK SECTION 3.1.**

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1. let S be the set of vertices saturated by a matching M in a graph G . Prove that some maximum matching also saturates all of S . (Hint: Use Berge's Theorem to show that one can start with any matching M , and find a sequence of "alternating path augmentations" that produce a maximum matching M^* . Be careful not to conflate "maximum" with "maximal.")

Proof. Start with $M_0 = M$. If M_i is not maximum, Berge's Theorem gives an M_i -augmenting path P : its edges alternate between edges outside and inside M_i , and its two endpoints are unsaturated by M_i . Set

$$M_{i+1} = M_i \triangle E(P).$$

This is a matching with $|M_{i+1}| = |M_i| + 1$. Every internal vertex of P loses one matching edge and gains one, the two endpoints become saturated, and vertices outside P retain their matching status. Thus every vertex saturated by M_i is also saturated by M_{i+1} .

Since a matching has at most $\lfloor |V(G)|/2 \rfloor$ edges, this process terminates. At termination there is no augmenting path, so Berge's Theorem implies that the resulting matching M^* is maximum. Saturation is preserved at every step; hence M^* saturates all of S . $\square$

2. Determine the minimum size of a maximal matching in the cycle C_n ,

Solution. For every $n \geq 3$, the minimum size is

$$\left\lceil \frac{n}{3} \right\rceil.$$

A matching is maximal precisely when no edge has two unsaturated endpoints: such an edge could be added, and any edge that could be added would have two unsaturated endpoints. Consequently, every edge of C_n either belongs to a maximal matching M or shares an endpoint with an edge of M . Each matching edge accounts for at most three cycle edges (itself and its two neighboring edges). Therefore $n \leq 3|M|$, giving the lower bound $|M| \geq \lceil n/3 \rceil$.

For a construction attaining this bound, label the vertices $v_1, \dots, v_n$ in cyclic order, write $n = 3q + r$ with $r \in \{0, 1, 2\}$, and start with

$$M_0 = \{v_{3j+1}v_{3j+2} : 0 \leq j \leq q-1\}.$$

If $r = 0$, take $M = M_0$. If $r = 1$, take $M = M_0 \cup \{v_{3q}v_{3q+1}\}$. If $r = 2$, take $M = M_0 \cup \{v_{3q+1}v_{3q+2}\}$. In every case these edges are pairwise disjoint and the unsaturated vertices are pairwise nonadjacent, including across the edge v_nv_1 . Thus M is maximal. Its size is q when $r = 0$ and $q + 1$ otherwise, as required. $\square$

3. Let M and M' be matchings in an X, Y -bigraph G . Suppose that M saturates $S \subseteq X$ and M' saturates $T \subseteq Y$. Prove that G has a matching that saturates $S \cup T$

Proof. Consider the spanning subgraph H with edge set $M \cup M'$. Every vertex has degree at most two. Each edge in $M \cap M'$ is an isolated edge component. Every other nontrivial component is a path or an even cycle whose edges alternate between M and M' .

On a cycle, either set of alternating edges saturates every vertex. On an isolated common edge, choose that edge. On a path component, every internal vertex is saturated by both matchings, so only its endpoints can require a choice. An endpoint in S can require the M -edges, and an endpoint in T can require the M' -edges.

These requirements cannot conflict on a single path. Indeed, a conflict would give an endpoint $s \in S \subseteq X$ incident only with an M -edge and another endpoint $t \in T \subseteq Y$ incident only with an M' -edge. The alternating path from s to t would then have even length, since its first and last edges belong to different matchings. An even-length path in a bipartite graph has its endpoints in the same part, contradicting $s \in X$ and $t \in Y$.

On each path choose the set of alternating edges satisfying its endpoint requirements, choosing either set if there are none. Isolated vertices belong to neither S nor T . The union of all chosen edges is a matching, since distinct components are vertex-disjoint, and it saturates $S \cup T$. $\square$

4. When $k \geq 2$, prove that Q_k has at least $2^{2^{k-2}}$ perfect matchings

Proof. Represent the vertices of Q_k by binary strings of length k , with two vertices adjacent when they differ in exactly one coordinate. Fixing the last $k-2$ coordinates and varying the first two gives a copy of Q_2 , that is, a 4-cycle. The 2^{k-2} possible fixed strings partition $V(Q_k)$ into 2^{k-2} vertex-disjoint 4-cycles.

Each of these cycles has two perfect matchings. Choose one independently in each cycle and take their union. The union is a perfect matching of Q_k , since it saturates every vertex exactly once. Different choices give different perfect matchings: a change on any one cycle changes the edges of the union. Hence Q_k has at least

$$2^{2^{k-2}}$$

perfect matchings. This also covers $k = 2$, when the fixed string is empty and there is a single 4-cycle. $\square$

5. Let G be the Petersen Graph.

- (a) Prove that deleting any perfect matching of G leaves $C_5 + C_5$ (i.e. two disjoint 5-cycles). (Hint: After you remove a perfect matching what is the degree of each vertex).
- (b) Prove that the Petersen graph has exactly six perfect matchings. (Hint: use (a) and use the fact that you already know how many 5-cycles G contains.)

Proof. Use the standard labeling of the Petersen graph with outer vertices u_i and inner vertices v_i , where $i \in \mathbb{Z}/5\mathbb{Z}$, and edges

$$u_i u_{i+1}, \quad u_i v_i, \quad v_i v_{i+2}.$$

Call the edges $u_i v_i$ spokes.

(a) Let F be any perfect matching. Since the Petersen graph is 3-regular, $G - F$ is 2-regular and thus a disjoint union of cycles. We determine these cycles by considering the number r of spokes in F . The $5 - r$ outer vertices not matched by spokes must be paired by outer edges, so $5 - r$ is even. Hence $r \in \{1, 3, 5\}$.

The case $r = 3$ is impossible. If a, b are the two indices without matching spokes, then both $u_a u_b$ and $v_a v_b$ would have to be matching edges. The first requires $b - a \equiv \pm 1 \pmod{5}$, while the second requires $b - a \equiv \pm 2 \pmod{5}$, a contradiction.

If $r = 5$, deleting F leaves the outer 5-cycle and the inner 5-cycle, as desired.

If $r = 1$, rotational symmetry allows us to assume that the sole matching spoke is $u_0 v_0$. Deleting u_0 from the outer cycle leaves the path $u_1 u_2 u_3 u_4$, whose unique perfect matching is $\{u_1 u_2, u_3 u_4\}$. Deleting v_0 from the inner cycle leaves the path $v_2 v_4 v_1 v_3$, whose unique perfect matching is $\{v_2 v_4, v_1 v_3\}$. Thus

$$F = \{u_0 v_0, u_1 u_2, u_3 u_4, v_2 v_4, v_1 v_3\}.$$

Its deletion leaves precisely the two cycles

$$u_0 u_1 v_1 v_4 u_4 u_0 \quad \text{and} \quad u_2 u_3 v_3 v_0 v_2 u_2.$$

They are vertex-disjoint and together contain all ten vertices. This proves $G - F \cong C_5 + C_5$ in every case.

(b) The classification in part (a) also counts the perfect matchings. There is one with all five spokes and none with three spokes. For each of the five possible choices of a sole spoke, the two remaining 4-vertex paths force a unique perfect matching, as above. Each such choice does give a perfect matching, and the five are distinct because their sole spokes differ. Therefore the Petersen graph has exactly $1 + 5 = \boxed{6}$ perfect matchings. $\square$